

COMMERCIAL OFFICE ELECTRICAL DISTRIBUTION & CONTROLS DESIGN

Electrical Calculations & Design Basis

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AutoCAD Electrical

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Single-Storey Commercial Office Building

Electrical Systems:

600Y/347 V and 208Y/120 V

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Portfolio Design Project

This document presents the design basis and supporting electrical calculations for the accompanying E-101 through E-402 drawing set.

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DOCUMENT CONTROL

Document Title	Electrical Calculations & Design Basis
Project	Commercial Office Electrical Distribution & Controls Design
Building Type	Single-Storey Commercial Office
Drawing Set	E-101 through E-402
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1.0 DESIGN BASIS & SYSTEM OVERVIEW

1.1 Project Description

This project presents the electrical distribution and control design for a single-storey commercial office building. The design includes the main electrical distribution system, secondary distribution, lighting and receptacle systems, distribution panel loads, motor control, and packaged rooftop HVAC electrical connections.

The accompanying electrical drawing set consists of drawings E-101 through E-402. This calculation package documents the design basis, assumptions, electrical calculations, and equipment-selection rationale supporting the drawing set.

1.2 Electrical System Architecture

The building electrical distribution system is based on a 600Y/347 V, three-phase, four-wire main distribution system. MSB-1 distributes power to the building's major electrical loads, including secondary transformers, MCC-1, HVAC-1, HVAC-2, and DP-1.

Transformers T-1 and T-2 provide 208Y/120 V, three-phase, four-wire secondary power for LP-1 and PP-1 respectively. LP-1 supplies the building lighting loads, while PP-1 supplies receptacles and other 120/208 V power loads. DP-1 supplies 600 V heating equipment, MCC-1 supplies the building motor load, and dedicated 600 V feeders supply HVAC-1 and HVAC-2.

1.3 System Design Parameters

Parameter	Design Value	Drawing Reference
Main Distribution Voltage	600Y/347 V, 3Ø, 4W	E-101
Main Switchboard	MSB-1, 400 A	E-101
Secondary Distribution Voltage	208Y/120 V, 3Ø, 4W	E-101 / E-102
Lighting Panel	LP-1, 125 A	E-102 / E-301
Power Panel	PP-1, 200 A	E-102 / E-301
Distribution Panel	DP-1, 600 V, 100 A, 3Ø	E-101 / E-302
Motor Control Centre	MCC-1, 600 V, 100 A, 3Ø	E-101 / E-401
HVAC Equipment	HVAC-1 & HVAC-2, 600 V, 3Ø	E-101 / E-402

1.4 Calculation Conventions

Unless otherwise noted, electrical currents are calculated using the following relationships. Apparent power is expressed in volt-amperes (VA) or kilovolt-amperes (kVA), line-to-line voltage is represented by V_{LL} , and calculated current is represented by I .

Equation 1 — Single-Phase Apparent Power

$$I = \frac{S}{V}$$

Where:

I = calculated current (A)

S = apparent power (VA)

V = system voltage (V)

Equation 2 — Three-phase apparent power

$$I = \frac{S}{\sqrt{3}V_{LL}}$$

Where:

I = calculated line current (A)

S = three-phase apparent power (VA)

V_{LL} = line-to-line system voltage (V)

$\sqrt{3}$ = three-phase conversion factor, approximately 1.732

Equation 3 — Three-phase real power

$$I = \frac{P}{\sqrt{3}V_{LL}PF}$$

Where:

I = calculated line current (A)

P = three-phase real power (W)

V_{LL} = line-to-line system voltage (V)

PF = load power factor (dimensionless)

$\sqrt{3}$ = three-phase conversion factor, approximately 1.732

Where equipment-specific power factor, efficiency, minimum circuit ampacity (MCA), or maximum overcurrent protection (MOCP) data is required but no manufacturer data has been selected, the calculation will be identified as a design assumption rather than presented as manufacturer-verified data.

2.0 MAIN SERVICE & MSB-1 CALCULATIONS

2.1 MSB-1 Connected Loads

MSB-1 is the main 600Y/347 V distribution switchboard for the building. The switchboard supplies six outgoing feeders serving the lighting transformer, power transformer, motor control centre, two rooftop HVAC units, and DP-1. The connected downstream loads and equipment ratings are summarized below.

Feeder	Connected Equipment	System	Drawing Reference
FDR-1	T-1 / LP-1	600 V → 208Y/120 V	E-101 / E-102
FDR-2	T-2 / PP-1	600 V → 208Y/120 V	E-101 / E-102
FDR-3	MCC-1	600 V, 3Ø	E-101 / E-401
FDR-4	HVAC-1	600 V, 3Ø	E-101 / E-402
FDR-5	HVAC-2	600 V, 3Ø	E-101 / E-402
FDR-6	DP-1	600 V, 3Ø	E-101 / E-302

2.2 Known Downstream Connected Loads

The main service calculation begins with the connected loads established by the downstream panel schedules and equipment details. Where an equipment rating is known but an actual operating load has not been established, the equipment rating is not automatically treated as calculated building demand.

Load	Connected Load	Basis
LP-1	989 VA	E-301 connected load
PP-1	8,280 VA	E-301 connected load
DP-1	42,000 VA	E-302 connected load
MTR-1	15 hp	E-401 motor rating
HVAC-1	TBD	Manufacturer load data not selected
HVAC-2	TBD	Manufacturer load data not selected

2.3 Known Panel Load Subtotal

Before accounting for the motor and HVAC equipment, the known connected panel loads can be summed directly.

Connected Panel Load Formula

$$S_{total} = S_{LP1} + S_{PP1} + S_{DP1}$$

Where:

S_{total} = known connected-load subtotal (VA)

S_{LP1} = LP-1 connected load (VA)

S_{PP1} = PP-1 connected load (VA)

S_{DP1} = DP-1 connected load (VA)

Substitution:

$$S_{total} = 989 + 8280 + 42000$$

$$S_{total} = 51269 \text{ VA}$$

2.4 Equivalent Current of Known Loads

The equivalent 600 V three-phase current corresponding to the known 51.269 kVA connected-load subtotal is calculated using the three-phase apparent-power relationship.

Three-Phase Apparent Power Formula

$$I = \frac{S}{1.732 \times V_{LL}}$$

Where:

I = equivalent three-phase line current (A)

S = three-phase apparent power (VA)

V_{LL} = line-to-line system voltage (V)

1.732 = approximate value of $\sqrt{3}$

Substitution:

$$I = \frac{51269}{1.732 \times 600}$$

$$I = 49.3 \text{ A}$$

Equivalent Current of Known Loads = **49.3 A**

This number does not represent the final building service current because MTR-1 and the two HVAC units still have to be accounted for.

2.5 MSB-1 Rating Assessment

Drawing E-101 identifies MSB-1 as a 400 A, 600Y/347 V, three-phase, four-wire switchboard. The presently documented connected loads do not independently establish a requirement for a 400 A service. The 400 A rating therefore represents a project design selection that provides substantial capacity for HVAC demand, motor load, future expansion, and additional building loads not explicitly modeled in this portfolio design.

A final construction design would determine the required service rating using the applicable demand factors, continuous-load requirements, motor-load requirements, HVAC manufacturer data, and other requirements of the governing electrical code.

Design Selection: MSB-1 = 400 A, 600Y/347 V, 3Ø, 4W

3.0 TRANSFORMERS & SECONDARY PANELS

Transformers T-1 and T-2 step the 600 V distribution voltage down to 208Y/120 V for LP-1 and PP-1. The following calculations evaluate the connected secondary loads and corresponding transformer currents.

3.1 Connected Loads

Panel	Connected Load	Panel Rating	Drawing
LP-1	989 VA	125 A	E-301
PP-1	8,280 VA	200 A	E-301

Total Secondary Connected Load = 9,269 VA

3.2 T-1 / LP-1

LP-1 has a connected load of 989 VA. The equivalent transformer primary and secondary currents are calculated below.

LP-1 Secondary Connected Load Current Formula

$$I = \frac{S}{1.732 \times V_{LL}}$$

Where:

I = three-phase line current (A)

S = connected apparent power (VA)

VLL = line-to-line voltage (V)

$1.732 = \sqrt{3}$

Substitution:

$$I = \frac{989}{1.732 \times 208}$$

$$I = 2.75 \text{ A}$$

LP-1 connected-load current = 2.75 A at 208 V, 3Ø

T-1 Primary Connected Load Current Formula

$$I = \frac{989}{1.732 \times 600}$$

$$I = 0.95 \text{ A}$$

T-1 connected-load primary current = 0.95 A at 600 V, 3Ø

3.3 T-2 / PP-1

PP-1 has a connected load of 8,280 VA.

PP-1 Secondary Connected Load Current Formula

$$I = \frac{S}{1.732 \times V_{LL}}$$

Where:

I = three-phase line current (A)

S = connected apparent power (VA)

VLL = line-to-line voltage (V)

$1.732 = \sqrt{3}$

Substitution:

$$I = \frac{8280}{1.732 \times 208}$$

$$I = 22.98 \text{ A}$$

PP-1 connected-load current ≈ 23.0 A at 208 V, 3Ø

T-2 Primary Connected Load Current Formula

$$I = \frac{8280}{1.732 \times 600}$$

$$I = 7.97 \text{ A}$$

T-2 connected-load primary current ≈ 8.0 A at 600 V, 3Ø

3.4 Design Assessment

The calculated currents represent the presently modeled connected loads rather than the full capacity of LP-1 and PP-1. The higher panel and transformer capacities shown on the drawings provide allowance for spare circuits, future loads, and design growth. Final equipment sizing for construction would be verified using applicable demand factors and code requirements.

Parameter	LP-1 / T-1	PP-1 / T-2
Connected Load	0.989 kVA	8.280 kVA
208 V Current	2.75 A	23.0 A
600 V Primary Current	0.95 A	8.0 A
Panel Rating	125 A	200 A

4.0 DP-1 LOAD CALCULATIONS

DP-1 is a 600 V, three-phase distribution panel supplying three heating loads: EWH-1, UH-1, and UH-2. The calculations below determine the connected load and corresponding three-phase current. Refer to drawing E-302 for the DP-1 panel schedule and branch single-line diagram.

4.1 Connected Loads

Load	Description	Connected Load	Branch OCPD
EWH-1	Electric Water Heater	18 kW	30 A, 3P
UH-1	Unit Heater	12 kW	20 A, 3P
UH-2	Unit Heater	12 kW	20 A, 3P
TOTAL		42 kW	

Total DP-1 Connected Load = 42 kW

4.2 Individual Load Currents

For this portfolio calculation, the loads are treated as balanced three-phase resistive heating loads, so:

PF = 1.0

Three-Phase Line Current Formula

$$I = \frac{P}{1.732 \times VLL \times PF}$$

Where:

I = three-phase line current (A)

P = connected real power (W)

VLL = line-to-line voltage (V)

PF = power factor, assumed 1.0 for resistive heating

1.732 = approximate value of $\sqrt{3}$

EWH-1

Substitution:

$$I = \frac{18000}{1.732 \times 600 \times 1.0}$$

$$I = 17.32 A$$

EWH-1 calculated current = 17.3 A

UH-1 and UH-2

Because both unit heaters are identical 12 kW loads, one calculation applies to both:

$$I = \frac{12000}{1.732 \times 600 \times 1.0}$$
$$I = 11.55 \text{ A}$$

UH-1 calculated current = 11.5 A

UH-2 calculated current = 11.5 A

4.3 Total DP-1 Current

The total connected load is the sum of the three heating loads.

$$P_{total} = 18 + 12 + 12 = 42 \text{ kW}$$

$$I = \frac{42000}{1.732 \times 600 \times 1.0}$$
$$I = 40.41 \text{ A}$$

DP-1 connected-load current = 40.4 A

4.4 Design Summary

Parameter	Value
System Voltage	600 V, 3Ø
Connected Load	42 kW
Calculated Current	40.4 A
DP-1 Main Rating	100 A
EWB-1 Branch OCPD	30 A, 3P
UH-1 Branch OCPD	20 A, 3P
UH-2 Branch OCPD	20 A, 3P

The selected 100 A DP-1 provides capacity above the presently modeled 40.4 A connected load and allows for future expansion. Branch overcurrent protection shown on E-302 is treated as a project design selection; final conductor and overcurrent protection sizing would be verified against equipment nameplate data and applicable electrical code requirements.

5.0 MCC-1 MOTOR CALCULATIONS

MCC-1 supplies MTR-1, a 15 hp, 600 V, three-phase supply fan motor. Drawing E-401 shows the motor branch consisting of CB-M1, motor starter M1, overload relay OL-1, and MTR-1. The following calculation estimates the motor operating current for design evaluation.

5.1 Motor Input Current

Motor Output Power Formula

$$P_{out} = HP \times 746$$

Where:

P_{out} = motor mechanical output power (W)

HP = rated motor horsepower (hp)

746 = approximate watts per horsepower

$$P_{out} = 15 \times 746$$

$$P_{out} = 11190 \text{ W}$$

Motor mechanical output power = 11.19 kW

Motor Current Formula

$$I = \frac{P_{out}}{1.732 \times V_{LL} \times PF \times \eta}$$

Where:

I = estimated motor line current (A)

P_{out} = motor mechanical output power (W)

V_{LL} = line-to-line voltage (V)

PF = assumed motor power factor

η = assumed motor efficiency

1.732 = approximate value of $\sqrt{3}$

Assumed PF = 0.85

Assumed efficiency = 0.90

$$I = \frac{11190}{1.732 \times 600 \times 0.85 \times 0.90}$$

$$I = 14.1 \text{ A}$$

Estimated MTR-1 operating current ≈ 14.1 A

5.2 Motor Branch Design

Parameter	Design Value
Motor	MTR-1
Rating	15 hp
Voltage	600 V, 3Ø
Estimated Operating Current	14.1 A

Branch Breaker	CB-M1, 30 A, 3P
Starter	M1
Overload Relay	OL-1
Control Voltage	120 V from CPT-1

The 30 A branch breaker shown on E-401 is higher than the estimated normal operating current because motor branch overcurrent protection must accommodate motor starting current. OL-1 provides motor overload protection, while CB-M1 provides branch short-circuit and ground-fault protection. Final motor branch, overload, and conductor sizing would be verified using applicable motor tables, equipment data, and electrical code requirements.

5.3 Control Power

CPT-1 steps the 600 V MCC supply down to 120 V for the motor control circuit. The 120 V circuit operates the M1 starter coil and associated HAND/AUTO, START/STOP, BAS, RUN indication, and FAULT indication functions shown on E-401. A detailed CPT-1 VA calculation is not required because individual control-device coil and pilot-light burdens were not manufacturer-selected for this portfolio design.

6.0 HVAC ELECTRICAL DESIGN BASIS

HVAC-1 and HVAC-2 are rooftop HVAC units supplied from MSB-1 at 600 V, three-phase. Each unit is shown with a 60 A, 3-pole branch circuit and a local disconnecting means. Because manufacturer MCA and MOCB data were not selected for this portfolio design, the 60 A branch ratings are treated as project design selections rather than calculated equipment loads.

6.1 HVAC Distribution

Equipment	Source	Voltage	Branch OCPD	Disconnect
HVAC-1 / RTU-1	CB-4	600 V, 3Ø	60 A, 3P	DS-1, 60 A
HVAC-2 / RTU-2	CB-5	600 V, 3Ø	60 A, 3P	DS-2, 60 A

The branch circuit rating is not used as the actual HVAC operating current because rooftop unit current depends on manufacturer-selected compressors, fans, motors, heaters, and control equipment.

6.2 Maximum Circuit Capacity

Maximum Circuit Capacity Formula

$$S = 1.732 \times V_{LL} \times I$$

Where:

S = apparent power capacity (VA)

VLL = line-to-line voltage (V)

I = circuit current rating (A)
1.732 = approximate value of $\sqrt{3}$

$$S = 1.732 \times 600 \times 60$$

$$S = 62352 \text{ VA}$$

Maximum apparent-power capacity per 60 A branch $\approx$ 62.4 kVA

This value represents circuit capacity only and is not assumed to be the actual connected load of either HVAC unit.

6.3 Control Interface

Drawing E-402 shows BAS enable and space thermostat signals interfacing with the RTU control terminals. Internal compressor, fan, safety, and control circuitry is considered part of the packaged HVAC equipment and is therefore outside the detailed electrical design scope of this project.

Design Selection: HVAC-1 and HVAC-2 supplied at 600 V, 3Ø through individual 60 A branch circuits and local disconnects.

7.0 FEEDER SIZING & VOLTAGE DROP

Feeder and branch-circuit conductor sizing for a construction design depends on calculated load, conductor material, insulation rating, installation method, ambient conditions, and applicable electrical code requirements. Because detailed routing lengths and installation conditions were not established in this portfolio project, conductor sizes are not assigned as final construction values. The following calculation demonstrates the voltage-drop evaluation method using a representative feeder.

7.1 Representative Feeder

Parameter	Value
Load	DP-1
System Voltage	600 V, 3Ø
Connected Load	42 kW
Calculated Current	40.4 A
Panel Main Rating	100 A
Assumed One-Way Length	30 m

A 30 m one-way feeder length is assumed solely to demonstrate the voltage-drop calculation and does not represent a measured project routing distance.

7.2 Voltage Drop Calculation

Voltage Drop Formula

$$V_D = 1.732 \times I \times R \times L$$

Where:

V_D = calculated voltage drop (V)

I = load current (A)

R = conductor resistance per unit length (Ω/m)

L = one-way conductor length (m)

1.732 = approximate value of $\sqrt{3}$

Assumed conductor resistance: $R = 0.00082 \text{ } \Omega/\text{m}$

$$V_D = 1.732 \times 40.4 \times 0.00082 \times 30$$

$$V_D = 1.72 \text{ V}$$

Percent Voltage Drop Formula

$$V_{D\%} = \frac{V_D}{V_{LL}} \times 100$$

Where:

$V_{D\%}$ = percentage voltage drop

V_D = calculated voltage drop (V)

V_{LL} = nominal line-to-line voltage (V)

$$V_{D\%} = \frac{1.72}{600} \times 100$$

$$V_{D\%} = 0.29\%$$

Representative DP-1 feeder voltage drop $\approx 1.72 \text{ V}$, or 0.29% .

7.3 Feeder Design Assessment

The representative calculation demonstrates that feeder voltage drop depends on load current, conductor resistance, and routing length. Final conductor selection would require verification of allowable ampacity, conductor material, installation conditions, temperature correction, equipment termination requirements, and actual routing distance.

The conductor resistance and 30 m routing length used above are calculation assumptions only. They are included to demonstrate the design methodology and are not final construction selections.

8.0 FINAL DESIGN SUMMARY

The electrical distribution and controls design provides a coordinated single-storey commercial office system incorporating 600Y/347 V main distribution, 208Y/120 V secondary distribution, lighting and receptacle panels, mechanical loads, motor control, and HVAC power and control interfaces. The calculations in this document support the major design decisions shown throughout drawings E-101 through E-402.

8.1 Calculation Summary

System	Key Design Result
MSB-1	400 A, 600Y/347 V, 3Ø, 4W
LP-1	989 VA connected load
PP-1	8.280 kVA connected load
DP-1	42 kW / 40.4 A at 600 V, 3Ø
MTR-1	15 hp / estimated 14.1 A at 600 V, 3Ø
HVAC-1	600 V, 3Ø, 60 A branch
HVAC-2	600 V, 3Ø, 60 A branch
DP-1 Voltage Drop Example	1.72 V / 0.29%

8.2 Design Considerations

Equipment and distribution ratings were selected to provide a practical system architecture with capacity for the modeled loads and future expansion. Where manufacturer-specific data or final installation conditions were unavailable, assumptions were clearly identified rather than treated as construction values.

Final construction design would require verification against applicable electrical codes, equipment nameplate data, manufacturer requirements, actual conductor routing, installation conditions, available fault current, and coordination requirements.

8.3 Conclusion

The completed drawing and calculation package demonstrates the development of a commercial electrical distribution system from the main service through secondary distribution, lighting, receptacle loads, mechanical equipment, motor control, and HVAC interfaces. The design calculations provide engineering support for the major system loads and demonstrate three-phase power calculations, transformer loading, motor current estimation, panel loading, and voltage-drop analysis.

Electrical Calculation Package Complete — Refer to Drawings E-101 through E-402 for the corresponding design documentation.